AMITESH GUPTA

CONTACT

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Current city – Mumbai, India

Open to relocation – _______________

PROFILE

6 years combined research & applied experience in geospatial remote sensing and data analytics

17+ peer-reviewed publications

h-index 15 | 1,140+ citations

CORE COMPETENCIES

- Satellite remote sensing: Optical (Experienced). SAR, Hyperspectral & Thermal (Intermediate).
- Multi-scale data integration: flux tower, multi-satellite, reanalysis, climate models.
- Multivariate statistical modelling for causal inference (Transfer Entropy, PCMCI, Granger)
- ML-DL for spatial (image) classification and scale-variant temporal prediction framework.
- Google Earth Engine for data curation, process, analysis and visualisation through API.
- Procedural programming for big-data analysis: Modular Pipeline and parallel processing.
- Object-oriented programming: Encapsulation, Graph-based and Inheritance.
- Mentoring & project supervision

TECHNICAL SKILLS

Programming

Python (advanced), R (intermediate), Julia

Python Libraries

Xarray, Rasterio, Dask, PyArrow, SciPy, NumPy, hPy, Statsmodels, Scikit-learn, Pandas, GeoPandas, Optuna, Keras, PyTorch, shutil, gc, os, json.

GIS / Remote Sensing Software

QGIS, SNAP, PolSARPro, GrADS, Panoply

Satellite dataset

MODIS, VIIRS, AVHRR, AMSR, SMAP, OCO, ECOSTRESS, Landsat, Sentinel, OMI, AIRS

Climate dataset

ERA5, GLEAM, CMIP6, GPCC, CRU, HadCRUT, MSWEP, GISTEMP, IMERG, PERSIANN-CCS.

EDUCATION

Ph.D., Ecosystem–Climate Interactions

Indian Institute of Technology Bombay, 2021–2026

M.Tech., Remote Sensing & GIS

Indian Institute of Remote Sensing, 2017–2019

M.Sc., Geography

Savitribai Phule Pune University, 2015–2017

GEOSPATIAL SCIENTIST

Remote Sensing of Terrestrial Ecosystem & Disaster Management

PROFESSIONAL SUMMARY

- Geospatial researcher with 06 years of experience in satellite-based remote sensing, geospatial analytics in open source platforms, and machine-learning applications for forecasting and image classification.
- Efficient in operating with multi-temporal and multi-channel satellite data using big-data pipelines in Python; data curation and classification in Google Earth Engine.
- Experience in implementing several Machine Learning and Deep Learning algorithms (Random Forest, XGBoost, Support Vector Machine, Artificial Neural network) and familiar with data driven timeseries forecasting using LSTM, ARIMA, GARCH.
- Collaborative teamwork for several peer-reviewed research publications in various international journals. Mentored research interns for dissertation projects.

  Seeking to apply this foundation to crop monitoring, acreage estimation, and yield forecasting for resilient farming systems in the semi-arid tropics.

EXPERIENCE

Doctoral Researcher — Terrestrial Ecosystem-Climate Interactions

Centre for Studies in Resources Engineering (CSRE), IIT Bombay | 2021 – 2026

- Designed and executed pipelines for simultaneous processing of multiple datasets, which includes ***satellite observations*** (MODIS, AVHRR, MSWEP, SMAP, EnMAP, ECOSTRESS, AMSR-E,2, OCO-2,3, Landsat-5,7,8,9, Sentinel-1/2,5p), ***reanalysis data*** (GLEAM, ERA5, GLDAS), ***climate model simulations*** (CMIP6) and ***ground observations*** (FLUXNET) to characterise global-scale ecosystem-climate interaction at varying temporal scales (climatological/seasonal/sub-seasonal/weekly/daily).
- Built statistical models to infer ***causal interaction*** among ecosystem productivity and water-use efficiency, canopy water content, biomass and land-atmosphere during the evolution of extreme dry events using Conditional Transfer Entropy, Multivariate Granger Causality and PCMCI, which are robust and transferable to other domains.
- Operated multi-source dataset in ***parallel processing***, resolving schema and memory-management issues in production-scale pipelines, which is comparable to the big-data/cloud-computing demands of operational environmental (or crop) monitoring.
- From doctoral thesis, three peer-reviewed research articles are published in different Q1 journals (Earth's Future, Geophysical Research Letters, Environmental Research Letters), as well as presented research works at IGARSS 2025, EGU 2025 & 2026.

Project: Burnt area detection using SAR

- Developed a Sentinel-1 band ratio-based classification workflow to detect burnt area amidst smoke, caused by wildfire in Mediterranean region, also tested over Canada.

Project: Inundated area mapping using SAR & Optical

- Developed an image classification workflow integrating Sentinel-1 SAR & Sentinel-2 spectral data to identify inundated/damaged area using stacked ML in Goa's estuary.

Project: Plant resilience estimation using eddy flux observations

- Developed a pipeline, based on long-term ground observations, to estimate resilience of different ecosystems from disturbance caused by drought or heatwave across USA.

Project: Drought recovery prediction using climate model simulations

- Developed a workflow to predict the rate of drought recovery in different biomes and their changes from the past to the future, using various climate model simulations.

GIS Analyst

GIS Cell, Dehradun, India | 2020 – 2021

- Conducted natural resource mapping, geodatabase preparation and potential zone identification for regional planning and development.
- Monitored disaster impact and forest disturbance using satellite-derived indicators, supporting field-validated geospatial assessments.

AWARDS

- ECS Travel Support, EGU 2026, Vienna
- IEEE Travel Grant, IGARSS 2025, Brisbane
- Gold Medallist, M.Sc. Geography

CONFERENCES

- IGARSS 2025, Brisbane, Australia
- EGU 2025 & 2026, Vienna, Austria
- ISPRS 2019, Dehradun, India

Research Visit

- Indiana University, Indianapolis, US (2026)
- Texas A&M University, Texas, US (2026)

REVIEW EXPERIENCE

Reviewer for 87+ manuscripts across 33 journals, including Nature Communications, Geophysical Research Letters, AGU Advances, Earth’s Future, Scientific Reports, Hydrological Sciences, Journal of Hydrology, Atmospheric Research etc.

LANGUAGES

English –Professional Proficiency

Hindi –Fluent Proficiency

Bengali –Native Proficiency

PUBLICATIONS

- **Gupta, A.**, Karthikeyan, L., … (2026). Dominant role of soil moisture regulating vegetation water content during flash drought evolution. Geophysical Research Letters, 53(8), e2025GL119366.
- **Gupta A.**, Saha S., … (2026): Intensifying compound dryness amplifies soil moisture control on water-use efficiency and vapor pressure deficit control on productivity across global terrestrial ecosystems. Environmental Research Letters (In Press)
- Khan, M. A., **Gupta, A.**, … (2026). Investigation of wildfire risk and its mapping using GIS-integrated AHP method: a case study over Hoshangabad Forest Division in Central India. Environment, Development and Sustainability, 28(4), 8681–8715.
- **Gupta A.**, Lanka L. (2025): Impact of Climate trends on Ecosystem Productivity. IGARSS 2025, Brisbane, Australia. doi:10.1109/IGARSS55030.2025.11243994
- **Gupta, A.**, & Karthikeyan, L. (2024). Role of initial conditions and meteorological drought in soil moisture drought propagation: An event-based causal analysis over South Asia. Earth's Future, 12(10), e2024EF004674.
- **Gupta, A.**, Roy, A., … (2023). Space-based observation of early summer wildfire event and its environmental proxies during 2021 in Eastern Peninsular India. Arabian Journal of Geosciences, 16(7), 433.
- Das, S., & **Gupta, A.** (2021). Multi-criteria decision based geospatial mapping of flood susceptibility and temporal hydro-geomorphic changes in the Subarnarekha basin, India. Geoscience Frontiers, 12(5), 101206.
- Bhatt, C. M., **Gupta, A.**, … (2021). Geospatial analysis of September 2019 floods in the lower Gangetic plains of Bihar using multi-temporal satellites and river gauge data. Geomatics, Natural Hazards and Risk, 12(1).
- **Gupta, A.**, Kant, Y., … (2021). Spatio-temporal distribution of INSAT-3D AOD derived particulate matter concentration over India. Atmospheric Pollution Research, 12(1).
- **Gupta, A.**, Banerjee, S., … (2020). Significance of geographical factors to the COVID-19 outbreak in India. Modeling Earth Systems and Environment, 6(4).
- Rousta, I., … **Gupta, A.**, (2020). Impacts of drought on vegetation assessed by vegetation indices and meteorological factors in Afghanistan. Remote Sensing, 12.
- Prabhu, V., … **Gupta, A.**, … (2020). Black carbon and biomass burning associated high pollution episodes observed at Doon Valley in the foothills of the Himalayas. Atmospheric Research, 243, 105001.
- **Gupta, A.**, Pradhan, B., ... (2020). Estimating the impact of daily weather on the temporal pattern of COVID-19 outbreak in India. Earth Systems and Environment, 4(3).
- **Gupta, A.**, Moniruzzaman, M., … (2020). Estimation of particulate matter (PM2.5, PM10) concentration and its variation over urban sites in Bangladesh. SN Applied Sciences, 2(12).
- **Gupta, A.**, Bhatt, C. M., … (2020). COVID-19 lockdown — a window of opportunity to understand the role of human activity on forest fire incidences in the Western Himalaya, India. Current Science, 119(2).
- Jana, S., **Gupta, A.**, … (2020). Assessment of global performance on COVID-19 research during 1990–2019: an exploratory scientometric analysis. Qualitative and Quantitative Methods in Libraries.
- Das, S., **Gupta, A.**, … (2017). Exploring groundwater potential zones using MIF technique in semi-arid region: a case study of Hingoli district, Maharashtra. Spatial Information Research, 25(6).